Product Sales Forecasting

Project submission · **Guntuku Hariprasad** · GitHub @dot-hpg

Daily sales forecasts for 365 retail stores, 61 days ahead (1 June to 31 July 2019), with a global LightGBM model deployed as a mobile-friendly web app and a Flask REST API.

Submission links

Deliverable	Link	Contents
GitHub repository	github.com/dot-hpg/sales-forecasting	Notebook, code, README, tests, processed data
Jupyter notebook	github.com/dot-hpg/sales-forecasting/blob/main/notebooks/Product_Sales_Forecasting.ipynb	EDA, hypothesis tests, 10 models, tuning, SHAP, insights
Live web app	To be added	Streamlit forecasting app (works on mobile)
REST API	github.com/dot-hpg/sales-forecasting/blob/main/api/app.py	Flask API source, Dockerfile and Render blueprint
Tableau dashboard	To be added	Tableau Public: sales, promotions, holidays, forecast accuracy
Technical blog	To be added	About 2,000-word write-up of the project
Demo video	To be added	5-minute Loom walkthrough
Portfolio	To be added	datascienceportfol.io profile (optional)

Problem and approach

A retail chain needs a daily sales forecast per store for the next two months to plan inventory, budgets, promotions and staffing. The data covers 365 stores (4 store types, 5 location types, 4 regions) and 516 days (188,340 rows, January 2018 to May 2019) with holiday and discount flags, orders and sales.

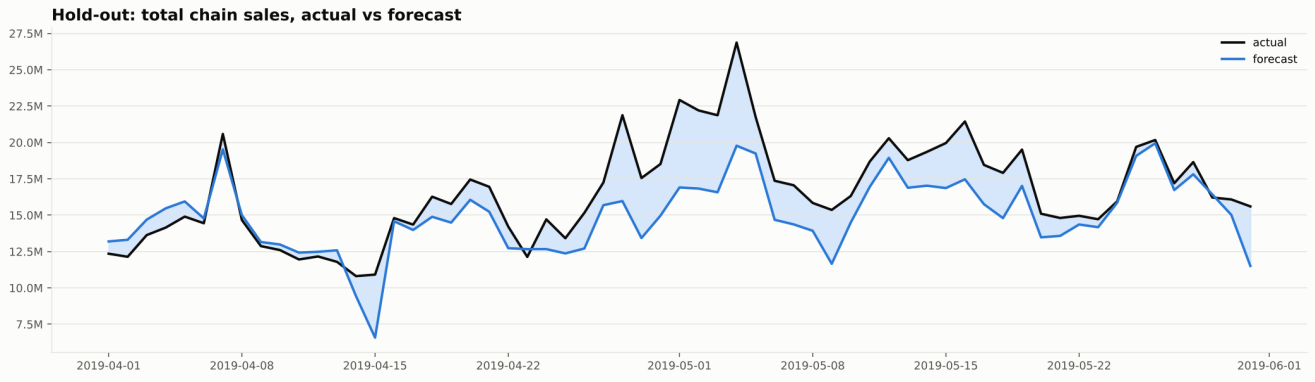
Pipeline: data quality checks → EDA → 8 hypothesis tests with effect sizes → leakage-free feature engineering → rolling-origin cross-validation (three 61-day folds) → 10 models compared → final hold-out test → Streamlit app and Flask API. **#Order is excluded** because it is only known after the day ends (target leakage), and rolling sales features are shifted by the full 61-day horizon.

Key results

Result	Value
Cross-validation WAPE: LightGBM vs seasonal naive	**15.3%** vs 22.8%
Chain-level WAPE: LightGBM vs SARIMAX / Prophet / Holt-Winters	**10.4%** vs 14.2 / 17.2 / 18.1%
Hold-out Apr-May 2019, store × day WAPE (R^2)	**16.8%** (0.67)
Hold-out Apr-May 2019, store × 61-day total WAPE (R^2)	**10.5%** (0.83)
Forecast for June and July 2019, whole chain	**989.8M** (−2.6% vs 2018)

Insights and recommendations

Finding	Evidence	Recommendation
Discounts lift sales	+33.5% per store, 364 of 365 stores	Run campaigns on weaker days (Tue to Fri) and months
Holidays reduce sales	−19%; −48% from day 2 of long holidays	Cut staff and fresh stock, especially on long holidays
Weekends peak	Sunday +15%, Saturday +9%	Staff up and replenish before weekends
Format and location set the level	S4 59.9K vs S2 27.5K per day; region $\eta^2 = 0.03$	Prioritise S4/S3 formats in L2/L1 locations
Forecast drifts with record months	Hold-out bias −9.8% after record May 2019	Retrain monthly and alert if bias exceeds ±5%



Hold-out test: daily chain sales, actual vs 61-day-ahead forecast (April to May 2019).

Brief coverage

Brief requirement	Where it is
Tableau visualisation	Tableau Public dashboard (link above); data in data/processed/
EDA and hypothesis testing	Notebook sections 3 to 7; data/processed/hypothesis_results.csv
ML modelling and time-series CV	Notebook sections 8 to 14; data/processed/model_comparison.csv
Deployment (serialised model + Flask API)	models/*.joblib, api/app.py, Dockerfile, streamlit_app.py
README, blog and demo video	README.md; blog and Loom links above